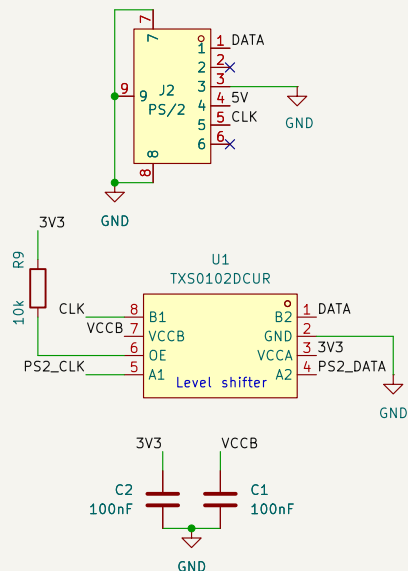
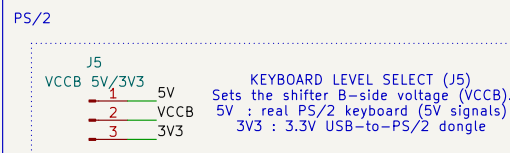
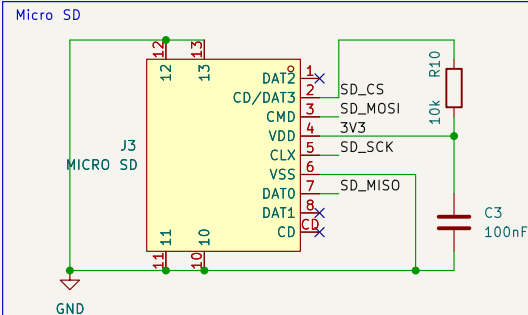
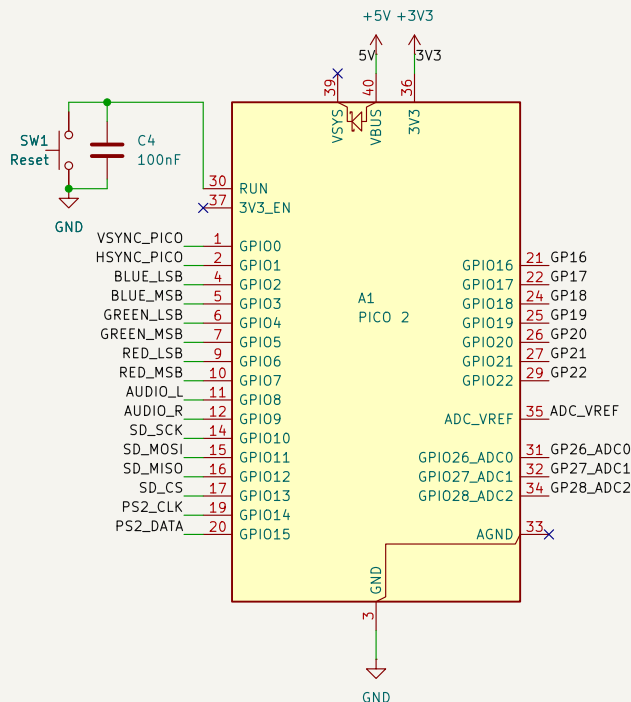
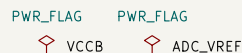




ERC POWER FLAGS
Mark externally-fed nets as driven
VCCB via J5 jumper, ADC_VREF via breakout
No electrical function.
Required to pass KiCad ERC



VGA:

VGA DAC – 2 bits/colour weighted resistors (470R MSB + 1k LSB),
100R series on H/V sync. 1024x768 @ 60Hz, 65 MHz dot clock.
3.3V sync drives standard monitors fine.

AUDIO:

PWM stereo + RC reconstruction filter per channel: 1k + 3n3,
10uF DC-block to powered speakers. C7/C8 polarised: + at filter node.

MICRO SD:

microSD over SPI1, 3.3V, no level shifter. R10 = CS pull-up.
Card-detect unused – its IRQ would disturb the VGA timing.
Bij PS/2 (naast de J5–nota):
TXS0102 bidirectional level shifter; internal pull-ups suffice.
J2 pin 4 stays 5V to power a USB-to-PS/2 dongle.

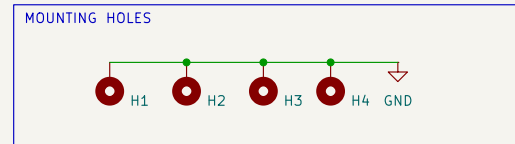
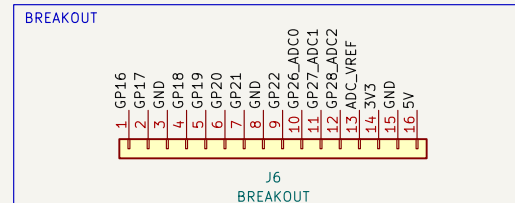
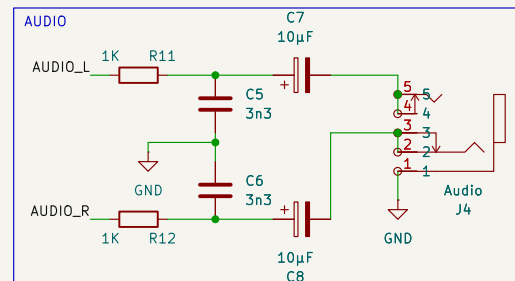
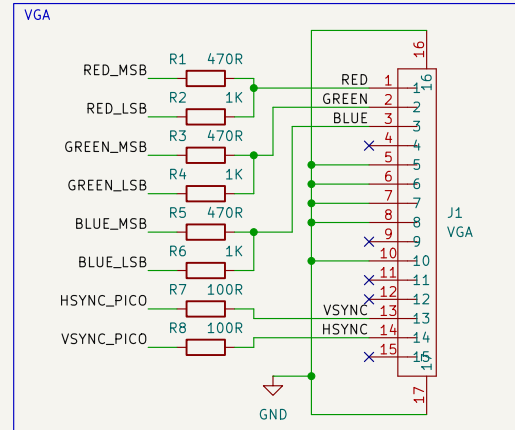
BREAKOUT:

Free GPIO for your own projects. GP26/27/28 are ADC inputs.
Feed an external reference on ADC_VREF if needed.

OTHER

4x M3 mounting holes, plated and tied to GND.

Powered from the Pico USB. 3V3 = Pico regulator (pin 36),
5V = VBUS (pin 40). VSYN, 3V3_EN and ADC_VREF left open.



<https://github.com/JanFromBelgium/japi-base>
License: GPL-3.0-or-later
VGA 1024x768, PS2 keyboard, microSD, stereo audio
Open retro computer platform – RP2350 (Pico 2)

Sheet: /
File: Japi_Base.kicad_sch

Title: Japi Base

Size: A4
Date: 2026-06-01
KiCad E.D.A. 10.0.3

Rev: June 1, 2026
Id: 1/1